Electrical Health & Safety

Engineering Technology

May, 2026

South East Technological University

Ollscoil Teicneolaíochta an Oirdheiscirt

Electrical Health & Safety (A17808)

Short Title: Electrical Health & Safety
Faculty Science and Computing
Department Engineering Technology
Cluster Engineering
Author CLAIRE.FITZPATRICK
Credits 5 **Level:** Intermediate

Description of Module / Aims

This module introduces the student to the various requirements for Electrical Engineers working in hazardous environments. The general legal principles of health and safety law are introduced along with relevant statutory obligations.

Programmes

BEng (Hons) in Electrical Engineering (`WD_ETRIC_B`)	3 / 5 / M
BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	3 / 5 / M
BEng (Hons) (`WD_ETRIC_B`)	3 / 5 / M

Indicative Content

- Work Practices in Hazardous Environments: General principles, ignition properties and apparatus protection.
- Standards, Directives, Practices and ATEX legislation.
- Relevant Statutory Bodies within the Electrical Engineering Sector.
- Legislation relating to overhead cables, underground cables, risk assessment, safety plans
- Health and safety duties under the common law, Principle of ``reasonable practicable''
- Duties of Project Supervisor Design Process (PSDP)

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Determine the selection and use of electrical apparatus in potentially explosive atmospheres.
2. Distinguish the Statutory Bodies within the Electrical Engineering Sector.
3. Determine the Health & Safety requirements in relation to the Electrical Engineering Sector.
4. Apply common law and legislative principles to the Health & Safety role of the electrical engineer.

Learning and Teaching Methods

- Lectures
- Tutorials
- Case Studies
- Presentations / Seminars

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,4
Continuous Assessment	50%	
Assignment	50%	2,3

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	48	
Independent Learning	87	

Essential Material

- E., T. _ETCI National Rules for Electrical Installations in potentially explosive atmosphere's_. 3rd. .: ET105, 2011.
- E., T. _ETCI National Rules for Electrical Installations_. 4th . .: ET101, 2008.
- McMahon, M., G. Shannon and M. Darlington. _A-Z of Health and Safety Risk Assessment in Ireland_. .: Round Hall, 1998.
- Shannon, G. _Health and Safety: Law and Practice_. .: Round Hall, 2007.

Supplementary Material

- "www.atex.ie." www.atex.ie
- "www.bailii.org." www.bailii.org
- "www.etci.ie." www.etci.ie
- "www.hsa.ie." www.hsa.ie
- "www.oireachtas.ie." www.oireachtas.ie

- E., T. _Good practice Guide on the Management of Electrical Safety at Work_. .: ET206, 2009.
- E., T. _Guide to the basic principles of Electrical Safety_. .: ET2013, 2007.
- McMahon, M., G. Shannon and M. Darlington. _``A-Z of Health and Safety Risk Assessment in Ireland"_. .: Round Hall, 1998.

Requested Resources

- EQUIPMENT: MAC PCs
- ENGINEERING LAB: Electronics
- Lecture Room: Loose Seated